Course Name : Community Engagement Project
Course Code: BIT25437A0B

Project Title: Inclusive Crowdsourcing: A Multilingual, SMS/Voice-Based Framework for Bridging Accessibility Gaps in Disaster Data Collection

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Introduction:

- Disaster response increasingly relies on **crowdsourced data** — citizen reports, social media and volunteered geographic information (VGI).
- Most existing systems are built around **English, Twitter-centric, smartphone** inputs.
- Yet many residents of flood-prone areas speak **Marathi or Hindi**, use **basic phones**, or have low digital literacy — so they go unheard.
- The most vulnerable (elderly, rural, low-income) are often the **least represented** in the very data meant to help them.
- Our project builds an **inclusive, multilingual, SMS/voice-ready** reporting framework, implemented as the working prototype **AwaazSetu** for Pune.

Literature Survey & Findings : (Survey 1 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
1.	Unpacking the role of volunteered geographic information in disaster management: focus on data quality https://www.tandfonline.com/doi/full/10.1080/19475705.2023.2300825	Year: 2023 Taylor & Francis Review paper	• VGI and citizen science are related but distinct concepts • Combined bibliometric and text-mining methodology • Five thematic clusters in data-quality literature • VGI quality challenges mapped onto the PPRR disaster cycle
2.	A method for assessing the credibility of volunteered geographic information in case of flood crisis https://www.sciencedirect.com/science/article/pii/S1877050922011012	Year: 2022 ScienceDirect Technology paper	• Score-based credibility model for text- and map-based VGI • Uses distance, flood hazard and spatial clustering • Validated on Ushahidi reports from the 2013 Brisbane flood • Reports classed high/low credibility against ground truth
3.	Analysis of Mumbai floods in recent years with crowdsourced data https://www.sciencedirect.com/science/article/abs/pii/S2212095524000117	Year: 2024 ScienceDirect Technology paper	• Urban flooding drivers and impacts • Gaps in traditional flood monitoring • Crowdsourced data as a solution • Choice and challenges of Twitter as a source

Literature Survey & Findings : (Survey 2 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
1.	Opportunities for Crowdsourcing in Urban Flood Monitoring https://doi.org/10.1016/j.envsoft.2021.105124	2021 Environmental Modelling & Software (Elsevier)	• Reviews crowdsourced flood-monitoring sources • Examines webcams, social media and citizen science • Shows potential for fine-scale flood monitoring • Identifies data quality and integration challenges
2.	Integrating User-Generated Content to Track Urban Flooding Hotspots and Foster Emergency Management: A Case Study in Central China https://doi.org/10.1111/tgis.12836	2021 Transactions in GIS (Wiley)	• Uses UGC to identify flood hotspots • Applies crawlers, NLP and geographic visualisation • Validates with water level, DEM and rainfall • Supports flood emergency management
3.	Quality Assessment of Crowdsourced Social Media Data for Urban Flood Management https://doi.org/10.1016/j.compenvurbsys.2021.101690	2021 Computers, Environment and Urban Systems (Elsevier)	• Assesses quality of crowdsourced data • Uses flood-related Twitter data • Applies logistic regression and fuzzy logic • Estimates uncertainty in social media observations

Literature Survey & Findings : (Survey 3 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
1.	Deep Learning Methods for Flood Mapping https://hess.copernicus.org/articles/26/4345/2022/	26 Aug 2022 Hydrology and Earth System Sciences (Copernicus)	• Reviewed 58 studies using CNNs and RNNs • Faster detection of complex flood patterns • Poor performance in new regions without retraining • Transfer learning suggested for global use
2.	Machine Learning for Disaster Prediction and Preparedness https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2023.1194918/full	2 Nov 2023 Frontiers in Environmental Science	• Evaluated ML models for weather and flood prediction • ML detects complex patterns in climate data • Enables faster disaster early warnings • Supports scalable real-time alerts
3.	Comparison of Deep Learning Techniques for River Streamflow Forecasting https://ieeexplore.ieee.org/document/9423961	5 May 2021 IEEE Access, Vol. 9	• Compared six deep learning models • Daily discharge from seven Red River basin stations • LSTM and GRU outperformed feed-forward and CNN • Single-layer RNNs matched accuracy at lower cost

Literature Survey & Findings : (Survey 4 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
1.	A systematic review of social media-enabled flood disaster informatics: method, technology and application https://doi.org/10.1016/j.jag.2025.104935	2025 Int. J. of Applied Earth Observation and Geoinformation (Elsevier)	• Social media fills gaps in traditional monitoring • Multimodal data + AI boosts flood monitoring • Authenticity checks and integration remain open • Pipeline: analysis → detection → awareness → mapping → response
2.	Prototype citizen and community science app for urban heavy rainfall risk management in the Ahr Valley, Germany https://doi.org/10.5194/nhess-26-1795-2026	2026 Natural Hazards and Earth System Sciences (Copernicus)	• Personal risk perception drives action • Hybrid app → API → geo data warehouse • Privacy by design with anonymous IDs • Gaps: offline support, GNSS accuracy, validation
3.	Mobile and web-based application for flood data collection based on citizen science doi:10.1007/s12145-024-01664-1	2025 Earth Science Informatics (Springer)	• Modified Design Thinking suits small teams • App data + social images + satellite estimates • Structured DB enables web-GIS queries • CSV export; 97 users, 212 rows at prototype scale

Project Title:

Inclusive Crowdsourcing: A Multilingual, SMS/Voice-Based Framework for Bridging Accessibility Gaps in Disaster Data Collection

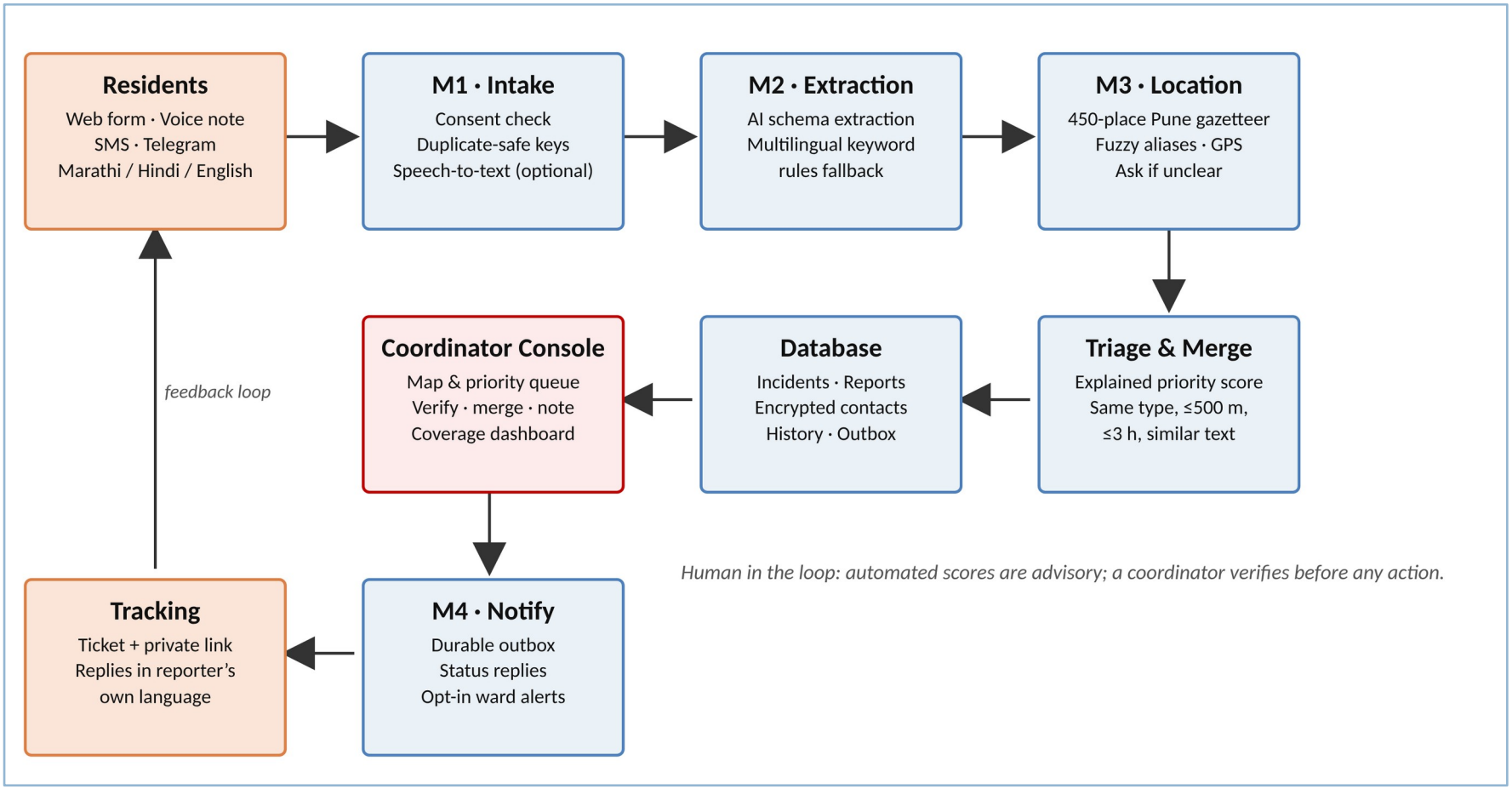
Working prototype: **AwaazSetu** (आवाज़ सेतु, "voice bridge") — a multilingual flood-reporting and coordination system for Pune community drills.

- Residents report by **web, voice, SMS or Telegram** in Marathi, Hindi or English.
- Coordinators see a **mapped, prioritised and explained** incident queue.
- A **coverage dashboard** shows which languages and areas are under-reported.

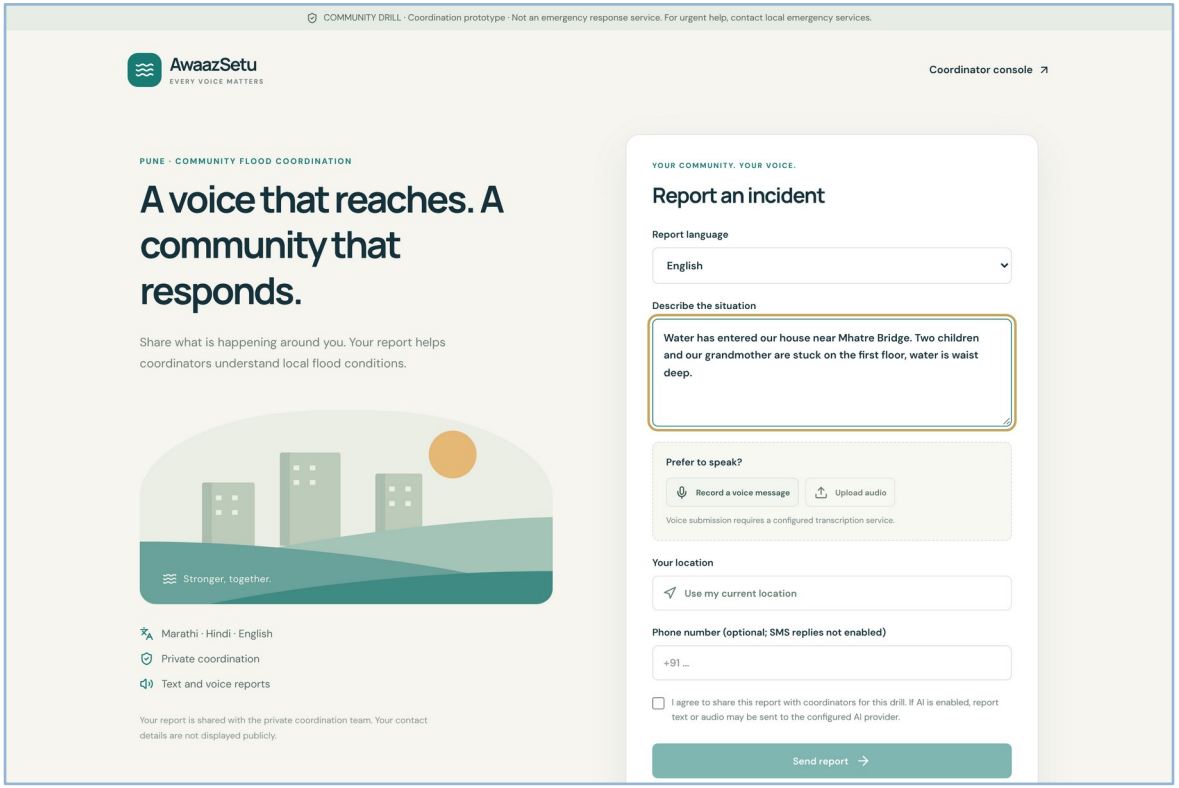
Objectives:

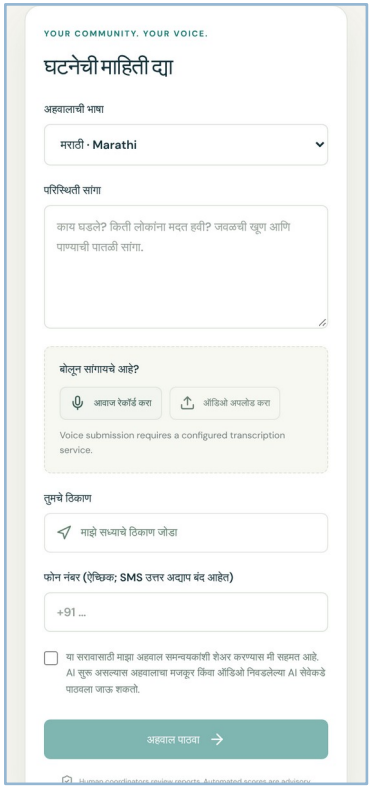
- Accept flood reports in **Marathi, Hindi and English** through **web, voice, SMS and Telegram**, so basic-phone users are included.
- Extract structured facts (incident type, people, water level, vulnerability) using **AI with a rules fallback** that works offline.
- Resolve informal **landmark names** to locations, and **ask instead of guessing** when unclear.
- Give each report a **transparent, explained priority score** and merge duplicates conservatively.
- Keep reporters informed in **their own language** and support **opt-in area alerts**.
- Measure **inclusion**: show which languages, channels and areas are under-represented.
- Protect **privacy** with consent, encrypted contacts and human review of every decision.

Proposed Methodology : (Block Diagram)



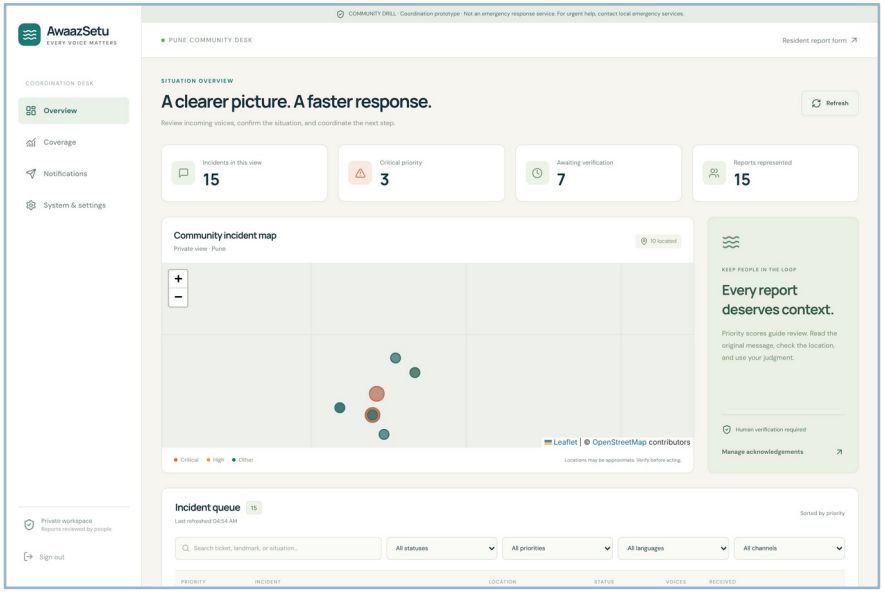
Implementation : Resident Interface

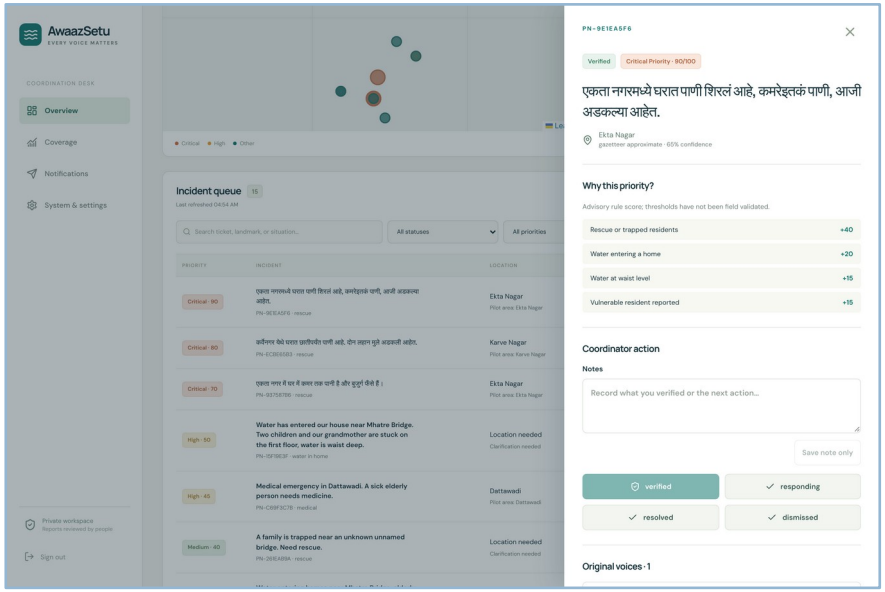




Features: three languages · voice with editable transcript · optional GPS · consent · ticket ID and private tracking link

Implementation : Coordinator Console





- **Overview:** live priority queue, private incident map, filters by status, priority, language and channel.
- **Incident detail:** score reasons, original message, verify / respond / resolve / dismiss, notes and merge with a reason.
- **Stack:** React + Vite, Leaflet, Recharts · FastAPI, SQLAlchemy, SQLite · optional OpenAI extraction and speech.

Result Analysis

Test / metric	Result	Notes
Backend automated tests	88 / 88 passed	pytest; API, triage, location, delivery, voice
Browser UI tests	10 / 10 passed	Playwright; resident and coordinator flows
Synthetic triage suite	150 / 150 exact match	50 cases each in EN, HI, MR; 7 fields
Report intake latency	median 7.4 ms, p95 9.8 ms	200 local reports, offline rules mode
Location lookup	0.24 ms per query	450-place Pune gazetteer
Field validation	Pending	Marathi speech accuracy, duplicate F1, landmark accuracy, SMS latency

Analysis: the rules engine handles all three languages consistently and responds in milliseconds, so it works offline and without AI cost. The synthetic suite is a regression check, **not field accuracy**; real accuracy needs a supervised community drill.

Conclusion:

- AwaazSetu shows that an **inclusive, multilingual, SMS- and voice-ready** channel can turn scattered flood reports into a **reviewed, located and prioritised** picture.
- Explained scores, conservative merging and **ask-don't-guess** locations keep coordinators in control.
- The coverage dashboard makes visible **who is still missing** from the data — the core gap found in the literature.
- **Future work:** connect the Android SMS gateway and Telegram bot, run a supervised drill, measure Marathi speech accuracy and duplicate F1, and add phone-ownership verification.

Thank You